Stability as Ground State — the Permanent Magnet as Proof

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.1 The Uncomfortable Question

Every textbook on quantum mechanics states: An electron exists in genuine superposition $\alpha|\uparrow\rangle + \beta|\downarrow\rangle$ — simultaneously in both spin states, until a measurement collapses the state.

On every refrigerator there is a permanent magnet.

Between these two facts lies a contradiction that standard theory never explicitly resolves.

The central question of this document

If a single, isolated electron exists in genuine superposition of both spin states — why do then billions of electrons in the permanent magnet *never* show collective superposition, but always unambiguous, stable alignment?
And conversely: What does the permanent magnet — an everyday object, not a laboratory experiment — tell us about the nature of spin?

.2 What the Permanent Magnet Really Shows

A permanent magnet is not a mysterious quantum phenomenon. It is the **macroscopic sum of spin alignments**.

Every electron in a ferromagnet (iron, nickel, cobalt) has a spin — and thus a magnetic moment. The **exchange interaction** (Doc. 174, section on Method 4) forces neighbouring spins into parallel alignment: all up, or all down. This is energetically more favourable than random distribution.

This collective alignment forms **Weiss domains** — macroscopic regions of coherent spin order. In the permanent magnet, these domains are permanently aligned and frozen by an external field.

What the permanent magnet directly proves

1. Spin has preferred, stable alignments.
Up and down are not equivalent alternatives — they are geometrically distinguished states. The field *wants* to be in one of them.

2. This stability is macroscopically robust.
A refrigerator magnet maintains its alignment at room temperature, in an environment with 10^{23} thermally fluctuating neighbouring atoms — for years, without decohering.
3. Superposition never occurs collectively.
No permanent magnet ever shows a state that is simultaneously north and south. Nature chooses. Always.

.3 Standard Theory and Its Explanation

Standard theory explains the macroscopic absence of superposition through **decoherence**: The interaction with the environment destroys superposition so quickly that it is practically unobservable.

This is technically correct — but it is a **post-hoc explanation**, not a prediction.

The structural weakness of the decoherence explanation

Standard theory postulates:

1. A single electron exists in genuine superposition.
2. Decoherence through the environment destroys this superposition practically immediately for macroscopic systems.

This does explain *why* we see no macroscopic superposition — but it reverses the physical order of events:
It begins with superposition as the ground state and then explains why this ground state is always immediately destroyed.
FFGFT reverses this: Stability is the ground state. Superposition is the limited, fragile exception — with measurable timescales and a geometric cause.
The question is not only mathematical — it is physical: which description is the more natural one?

.4 Superposition is Fragile — The Experimental Evidence

The measured coherence times from Doc. 174 tell an unambiguous story:

System	T_1	T_2	Temperature	Remark
Exciton (CdSe QD)	~ 1 ns	~ 10 ps	4 K	Superposition lasts picoseconds
Electron spin (GaAs)	~ 1 ms	~ 10 ns	mK	Superposition lasts nanoseconds
Electron spin (Si)	> 1 s	~ 10 ms	mK	Superposition lasts milliseconds
Nuclear spin (P:Si)	> 1 s	> 100 ms	mK	Best known system
Permanent magnet	**Years**	**—**	**300 K**	**Stability without cooling**

The table shows a clear trend: the more isolated the system from the disturbing influence of the environment, the longer superposition is maintained — but it *always* lasts only a limited time.

Superposition is never the stable ground state

Not a single experimental system shows superposition as a durably stable state. Even in the best qubit system (nuclear spin P:Si at mK) superposition decays after seconds.
The permanent magnet — without cooling, without isolation — maintains its stable spin state for years.
This is not coincidence. This is physics: **Stability is energetically preferred. Superposition is metastable.**

.5 The Single Electron: Superposition or Unknown State?

Here lies the actual conceptual core.

Standard theory claims: a single, isolated electron exists in genuine superposition $\alpha|\uparrow\rangle + \beta|\downarrow\rangle$ until measurement.

The FFGFT position is more precise:

FFGFT: Superposition vs. epistemic ignorance

A single electron has at every moment a **definite torsion configuration** — either spin-up or spin-down.

What we describe as superposition $\alpha|\uparrow\rangle + \beta|\downarrow\rangle$ is **epistemic ignorance** about this definitive state — not ontological simultaneity.
The proof comes from QND measurement (Doc. 174, Faraday rotation): a spin state can be measured repeatedly without changing it, and *always* yields the same result. This is only possible if the state *before measurement* exists definitively.
If the electron were truly in superposition, a QND measurement would yield random results at each repetition. It does not.

The apparent superposition arises in the description — when we mathematically describe a prepared system before the readout interaction. It is a state description under ignorance, not a physical double-state.

.6 The Bridge: From Single Spin to Permanent Magnet

The transition from quantum to classical magnetism is in FFGFT no puzzle — it is the same mechanism, only at different scales.

Level	**System**	**FFGFT description**	**Stability**
Single spin	Electron in QD	One stable torsion configuration; up or down	$T_1 > 1$ s (P:Si)
Spin pair	Two coupled QDs	Two torsions; exchange interaction prefers parallel alignment	$J(t)$ controllable
Domain	Weiss domain (10^{15} spins)	Collective torsion order; energetically more stable than disorder	Stable at 300 K
Permanent magnet	Macroscopic body	Frozen global torsion alignment	**Stable for years**

Each level is the same physics — stable torsion configurations, collectively reinforced by exchange interaction. No step in this chain requires genuine superposition. Every step requires **geometric precision and energetic stability**.

FFGFT: Permanent magnet as macroscopic resonator

In FFGFT the permanent magnet is not a special case — it is the natural macroscopic limiting case of the resonator principle from Doc. 173.
A single quantum dot at ξ-level $N \approx 8$: two stable torsion configurations, metastable intermediate states.
A permanent magnet: $\sim 10^{22}$ resonators forced by exchange interaction into the same stable torsion configuration. The collective field reinforces stability — instead of destroying it (decoherence), the coupling acts here as a stability amplifier.
The boundary between quantum mechanics and classical magnetism is in FFGFT no categorical boundary — it is a quantitative transition along the ξ-hierarchy.

.7 What Standard Theory Leaves Unanswered

The standard presentation of spin in textbooks and popular science texts has a structural problem: it begins with the abstract (Hilbert space, spinor, collapse) and then explains everyday life (permanent magnet) as a limiting case.

In doing so it overlooks a simple observation:

The simplest description of spin

The permanent magnet exists. It shows that spin is a **spatial orientation** — discrete, stable, real.
This is the most intuitive representation of spin: not an abstract state vector in Hilbert space, but a physical orientation in space that is macroscopically measurable, stable and everyday.
Quantisation (only up or down, nothing in between) follows from the geometry of the resonator — as derived in Doc. 173. It is not a postulate, but a geometric consequence.
The apparent superposition is an epistemic description under ignorance — not a physical double-state. The permanent magnet shows daily which state is fundamental: the stable, aligned one — not the superposed one.

.8 Consequences for Quantum Information

This view has direct consequences for the understanding of quantum computers.

The standard view: a qubit is powerful because it is in superposition. Decoherence is the enemy of superposition and must be combated.

The FFGFT view: a qubit is powerful because it is a **precisely controllable torsion configuration**. Decoherence is not the destruction of a miracle — it is the system returning to its energetically stable ground state. This is normal, not pathological.

Decoherence as return to the ground state

In FFGFT, decoherence is not a failure of the system — it is proof that the stable spin state is real and energetically preferred.

The permanent magnet does not decohere. The qubit decoheres in milliseconds. The difference is not the physics — the difference is the **isolation from the coupling** that enforces the stable alignment.

A quantum computer does not fight against the nature of spin — it fights against the spin's tendency to return to its natural stability before the computation is finished. This is an engineering problem, not a foundational problem.

.9 Summary

Conclusion: The Permanent Magnet as Witness

The permanent magnet is the simplest and most convincing argument for the FFGFT view of spin:

1. Spin has stable, preferred alignments.

Up and down are geometrically distinguished states — not arbitrary measurement outcomes.

2. Collective superposition does not exist.

Billions of electrons always choose one alignment. Never both simultaneously. Never.

3. Superposition is fragile — stability is robust.
Coherence times range from picoseconds (exciton) to seconds (nuclear spin) — always limited, always finite. The permanent magnet maintains its state for years.
4. The decoherence explanation reverses the physics.
Standard theory begins with superposition as the ground state and then explains why it always immediately disappears. FFGFT begins with stability as the ground state — superposition is the finite, measurably fragile exception.
5. QND measurement confirms determinism.
Repeated Faraday measurements always yield the same result — the spin state was definite *before* measurement.
In FFGFT language: Spin-up and spin-down are two stable torsion configurations. The permanent magnet is their macroscopic collective realisation. Superposition is a metastable intermediate torsion — real on short timescales, but never the fundamental ground state.
The simplest description of spin does not begin in Hilbert space — it begins at the refrigerator.

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